TRACKING SYSTEMSKYALERT:REALTIME FLIGHT

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CLASS DIAGRAM

USE CASE DIAGRAM

INTRODUCTION

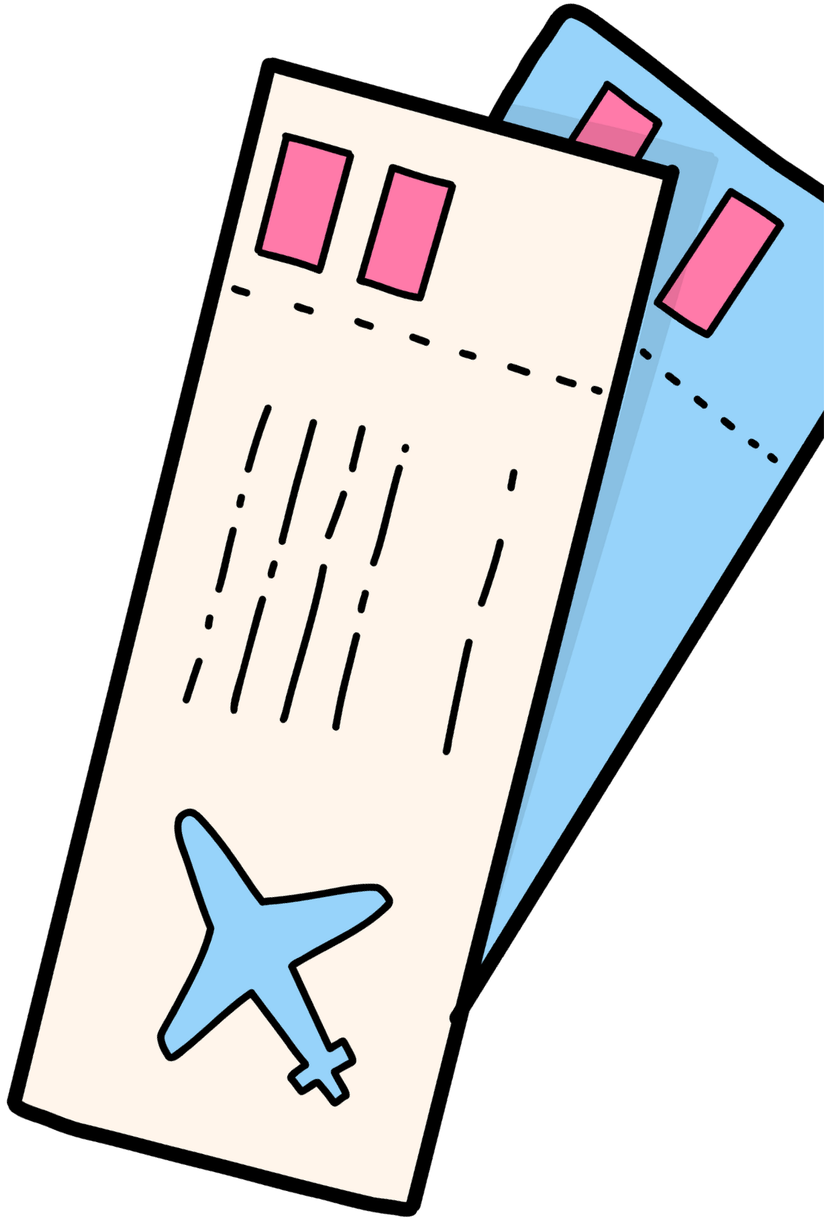


The Flight Alert System Using SMS is designed to improve communication between airports and passengers by delivering real-time flight updates directly to mobile phones. It sends important information such as flight delays, gate changes, and boarding status through SMS, without requiring internet access. This makes it especially useful for passengers with poor connectivity or limited mobile data.

By providing timely and accurate alerts, the system reduces confusion and ensures passengers stay informed throughout their journey, even in crowded or low-network airport environments.

The system also reduces the workload on airport staff by automatically sending updates without manual intervention. During peak travel hours, this improves efficiency and minimizes passenger inquiries regarding flight status.

Overall, the Flight Alert System enhances operational management, improves communication, increases customer satisfaction, and makes air travel more convenient and stress-free.

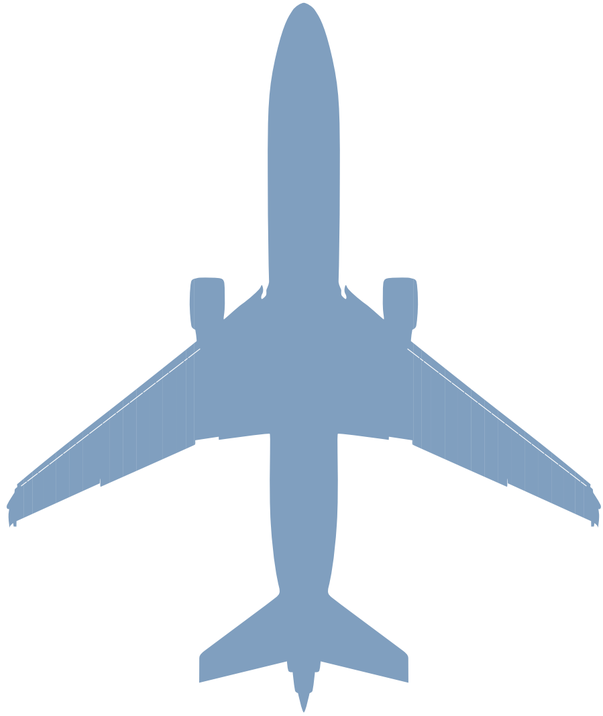


Literature Survey

1. SMS-Based Notification System for Transportation Services

Rahul Sharma, Neha Verma (2023)

- Explains how SMS alerts provide real-time updates like delays and schedule changes.
  - Shows that SMS is reliable and works even with weak internet. **Relevance:**
  - Supports the use of SMS for sending flight updates to passengers.



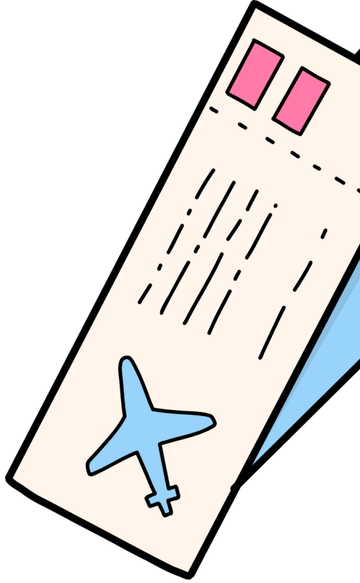
2. Design of Automated Alert Systems Using Mobile Communication

Priya Mehta, Karan Patel (2022)

- Describes systems that automatically send alerts using databases and messaging services.
- Helps reduce manual work and improve efficiency.
- **Relevance:** Helps understand automation of SMS alerts in the system

3. Real-Time Information Systems in Airport Management
Suresh Kumar, Anjali Rao (2024)

- Discusses how real-time communication improves airport operations and passenger experience.
- Suggests SMS as a reliable communication method.
- **Relevance:** Shows the importance of fast and reliable flight information.





4. Communication Technologies for Smart Transportation Systems
Rohit Malhotra, Sneha Iyer (2023)

- Compares communication methods like mobile apps, email, and SMS.
- Finds SMS more reliable in low-network areas.
- **Relevance:** Supports choosing SMS for the flight alert system.

Existing System:

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- **App-Based Solutions:** Require passengers to download apps and have internet access and storage space.

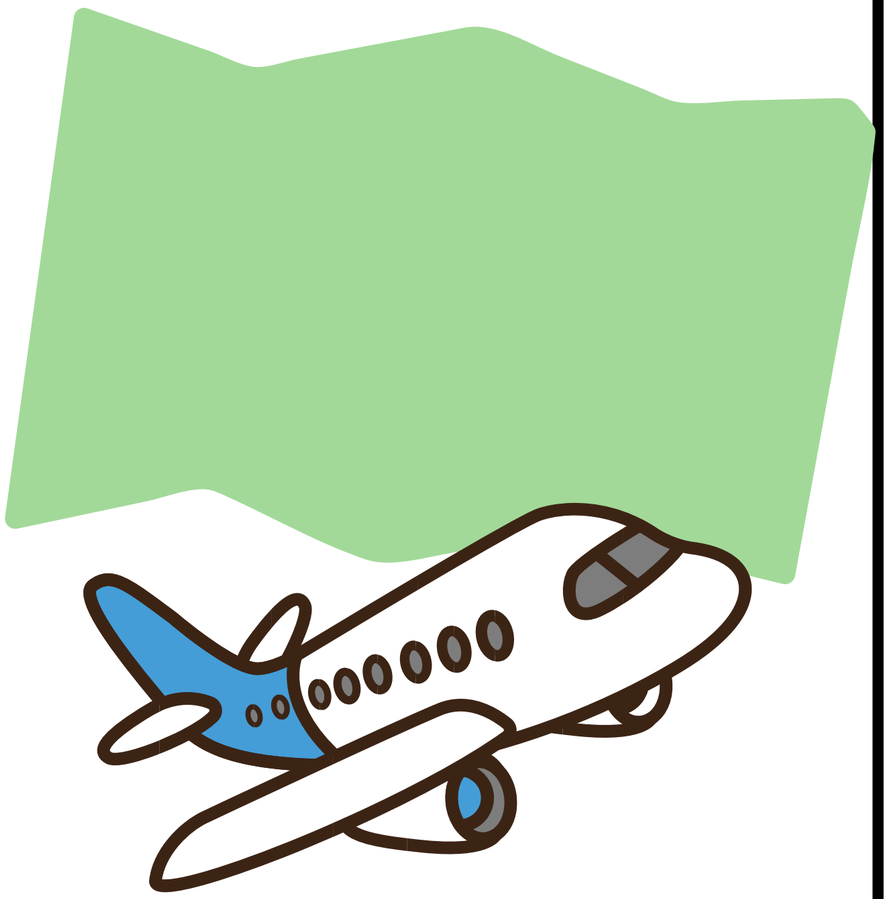
- **Limited Accessibility:** Passengers without mobile data, internet, or roaming access may miss important updates.

- **Crowded Information Desks:** Passengers often wait in long queues to get flight status or gate information.

- **Misheard Announcements:** Noise, unclear audio, or language barriers cause passengers to miss key updates.

requirements

software requirements



- Operating System: Windows 11
- Programming Language: Python
- Database: MySQL
- Front-End: Tkinter / Flask
- SMS API: Twilio
- Development Tool: VS Code / PyCharm
- Web Browser: Google Chrome / Microsoft Edge

requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB (Recommended: 8 GB)
- Hard Disk: Minimum 20 GB Free Space
- Computer / Laptop
- Internet Connection (For Admin System & SMS Setup)
- Mobile Phone (For Receiving SMS Alerts)

hardware requirements

purposed System

- **SMS-Based Communication:** Flight updates are sent directly via SMS without requiring internet or mobile apps .
- **Universal Accessibility:** Works for all passengers, including those without smartphones or mobile data .
- **Real-Time Updates:** Instant alerts on flight status, gate changes, boarding time, and delays .
- **Reduced Staff Workload:** Automates notifications, reducing passenger inquiries and staff burden.

working principle

1. Admin logs in securely to the dashboard

2. Admin adds/updates flight details in the system

3. Passengers are registered with their phone numbers$_i$

4. When a flight is delayed, cancelled or gate changed — admin sends an alert

5. Twilio API automatically sends SMS to all registered passengers 6. Systemlogs all sent/failed messages for tracking

MODULES

1. Admin Module: Allows authorized staff to securely login and manage flights, passengers, and alerts.

2. Flight Management Module: Stores and updates flight details like number, status, gate, and schedule in real time**.**

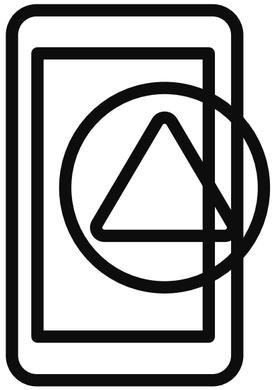
3. Passenger Management Module: Manages passenger records including phone numbers and flight assignments.

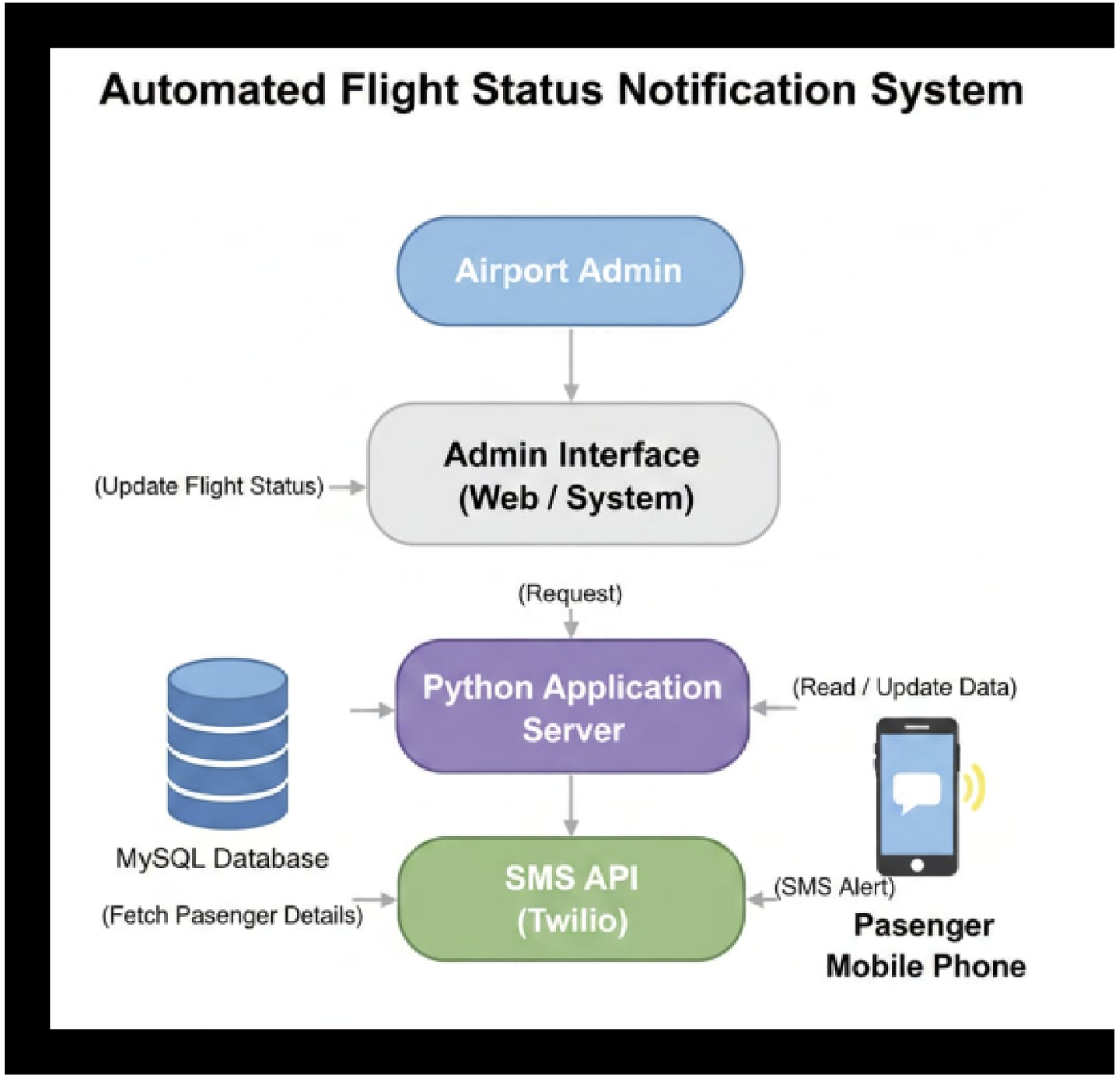
4. Database Module: Stores all flight, passenger, and SMS data securely using MySQL**.**

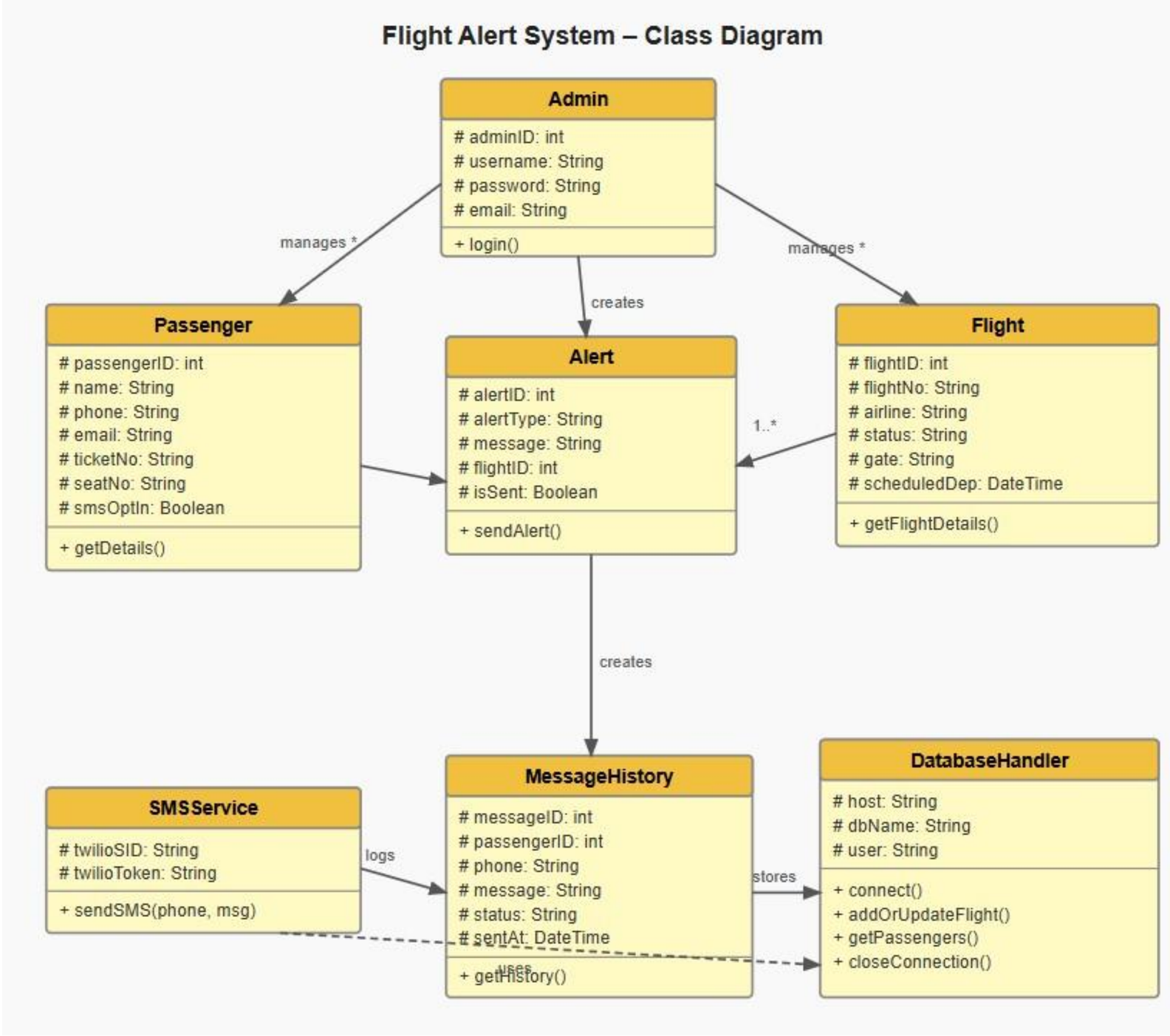
5. SMS Notification Module: Automatically sends flight alerts to passengers via Twilio SMS API

6. Frontend Module: Provides a responsive admin dashboard to manage the entire system easily**.**

system architecture



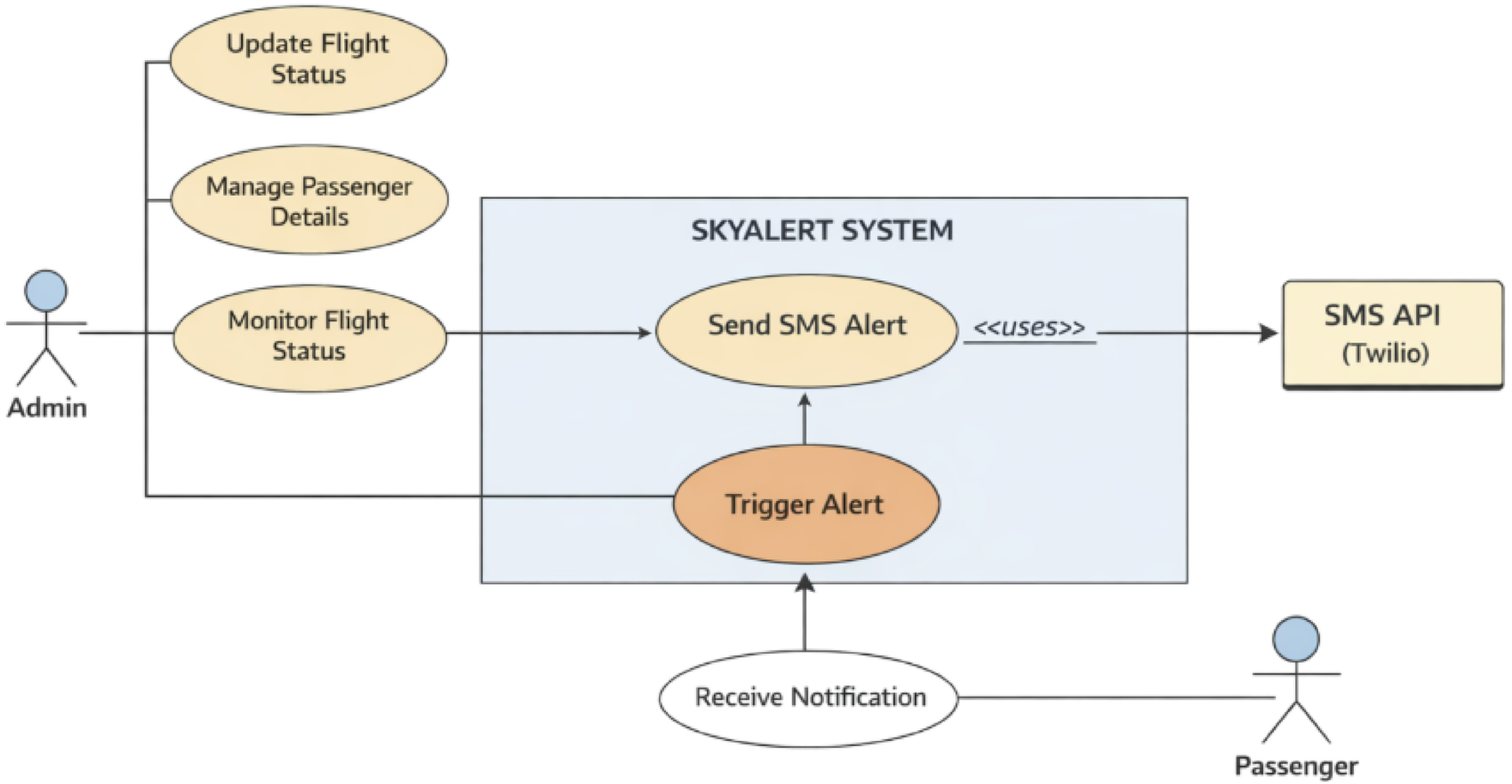




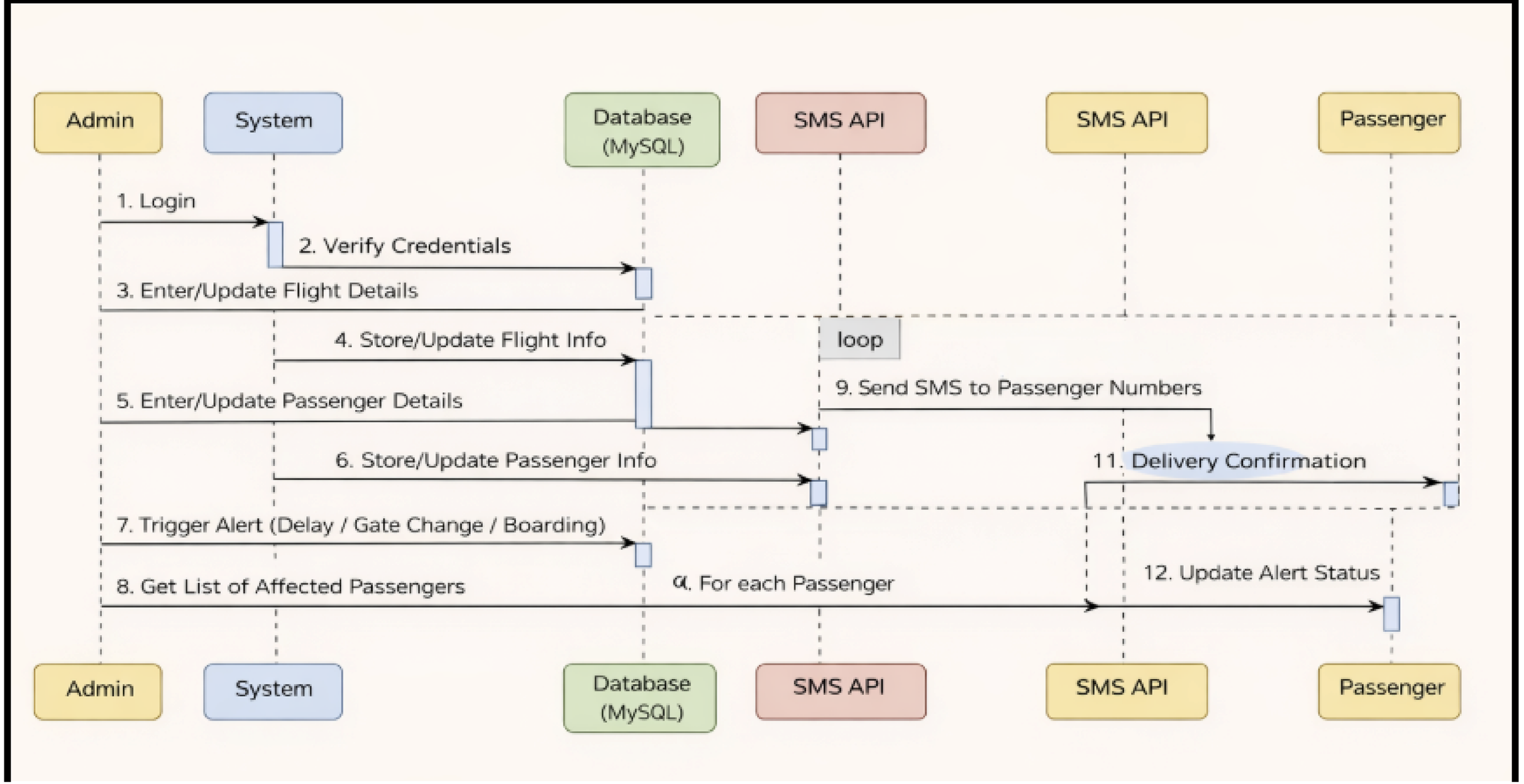
class DIAGRAM

USECASE DIAGRAM



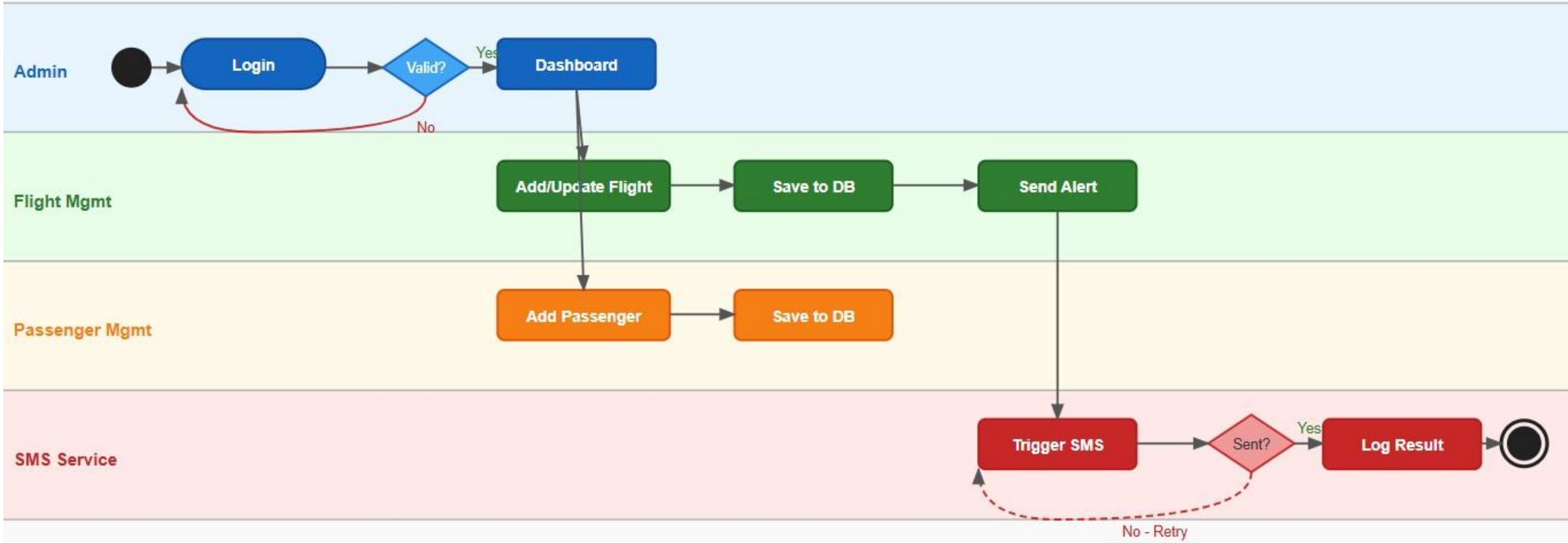


SEQUENCE DIAGRAM



activity DIAGRAM

Flight Alert System – Activity Diagram



conclusion



- The Flight Alert System using SMS provides an effective way to deliver important flight updates. • It reduces dependency on internet connectivity for passengers.
- Alerts such as delays, gate changes, and boarding information are sent on time.
- The system uses Python, MySQL, and SMS API for reliable communication.
- This project helps in reducing confusion and improves passenger experience.

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FUTURE ENHANCMENT





1. **Check-in Counter Alerts** Notify passengers automatically when check-in counters open for their flight.

2. **Connecting Flight Information**
Automatically send connecting flight details like gate, terminal and departure time to passengers.

THANK YOU



